

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26075
- **Problem Statement Title** – CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal to support organizational training, competency development, and knowledge sharing through a centralized web-based platform.
- **Theme** – Smart Education
- **PS Category** – Software
- **Team ID** –
- **Team Name (Registered on portal)** – console.log()



CAPACITY CONNECT

“ From fragmented training workflows to one connected capacity-building ecosystem “

BEFORE CAPACITY CONNECT

“ Learning resources, trainer expertise and trainee activity remain distributed across disconnected channels “



COURSE CONTENT

Courses & Modules
Presentations • study materials • recorded lectures



TRAINER INFORMATION

Trainer Profiles
Qualifications • skills • experience • expertise



ASSESSMENTS

Manual Assessments
MCQs • questionnaires • deadlines • evaluations



PARTICIPATION

Attendance & Participation
Limited visibility of learner engagement and progress



COMMUNICATION

Scattered Communication
Updates • reminders • instructions across channels



FEEDBACK

Feedback Collection
Learner feedback is difficult to consolidate and analyse



USER RECORDS

Disconnected Records
Profiles • enrollments • scores • certifications stored separately



TRAINER SELECTION

Who is the right trainer?
Expertise matching depends heavily on manual identification



ADMIN WORKLOAD

Manual Coordination
Approvals • monitoring • notifications • reporting take time

Multiple tools + manual coordination = fragmented learning experience

THE GAP

“ Disconnected information creates delays, blind spots and inefficient training decisions “



? HOW CAN WE RECOMMEND THE RIGHT COURSE FOR EACH TRAINEE?

Course prerequisites are often not connected with a trainee's skills, certifications, experience and domain knowledge, making it difficult to identify the right learning path..



? HOW CAN WE CONNECT TRAINING REQUIREMENTS WITH THE RIGHT TRAINER EXPERTISE?

No structured way to link **required competencies** with the **most suitable trainer expertise** for a training need.



? How can trainers create quality question papers faster?

Manual question-paper preparation is time-consuming and makes it difficult to generate topic- and difficulty-specific questions efficiently.



? How can trainees quickly understand and revise a module?

Long learning materials can make revision difficult, with no quick way to convert a module into concise personal notes.



? How can practice difficulty adapt to each trainee?

Fixed-difficulty quizzes do not respond to individual performance or continuously challenge learners at the right level.

AFTER CAPACITY CONNECT

“ One platform connecting trainees, trainers and administrators across the complete learning lifecycle “

LEARNING PATH ENGINE

Next-Best-Course Recommendation
Next module • gap-triggered • continued path • beyond completion

ADAPTIVE PRACTICE QUIZ

Dynamic Difficulty Adjustment
AI analyzes trainee performance and automatically adjusts question difficulty



COMPETENCY MAPPING

Weighted Trainer-Matching & Ranking
Skills & certificates • subject-wise mapping • eligibility filter • weighted score • top 3–5 suggestions • admin selection



One connected learning ecosystem



AI CONTENT STUDIO

AI-drafted question sets by topic & difficulty • trainer pattern-cloning • auto material summaries • Gemini course-fit recommendations



ASSESSMENT

Assessment with Effectiveness Score
Subject-wise MCQs • auto-scoring • feedback rollout • effectiveness score

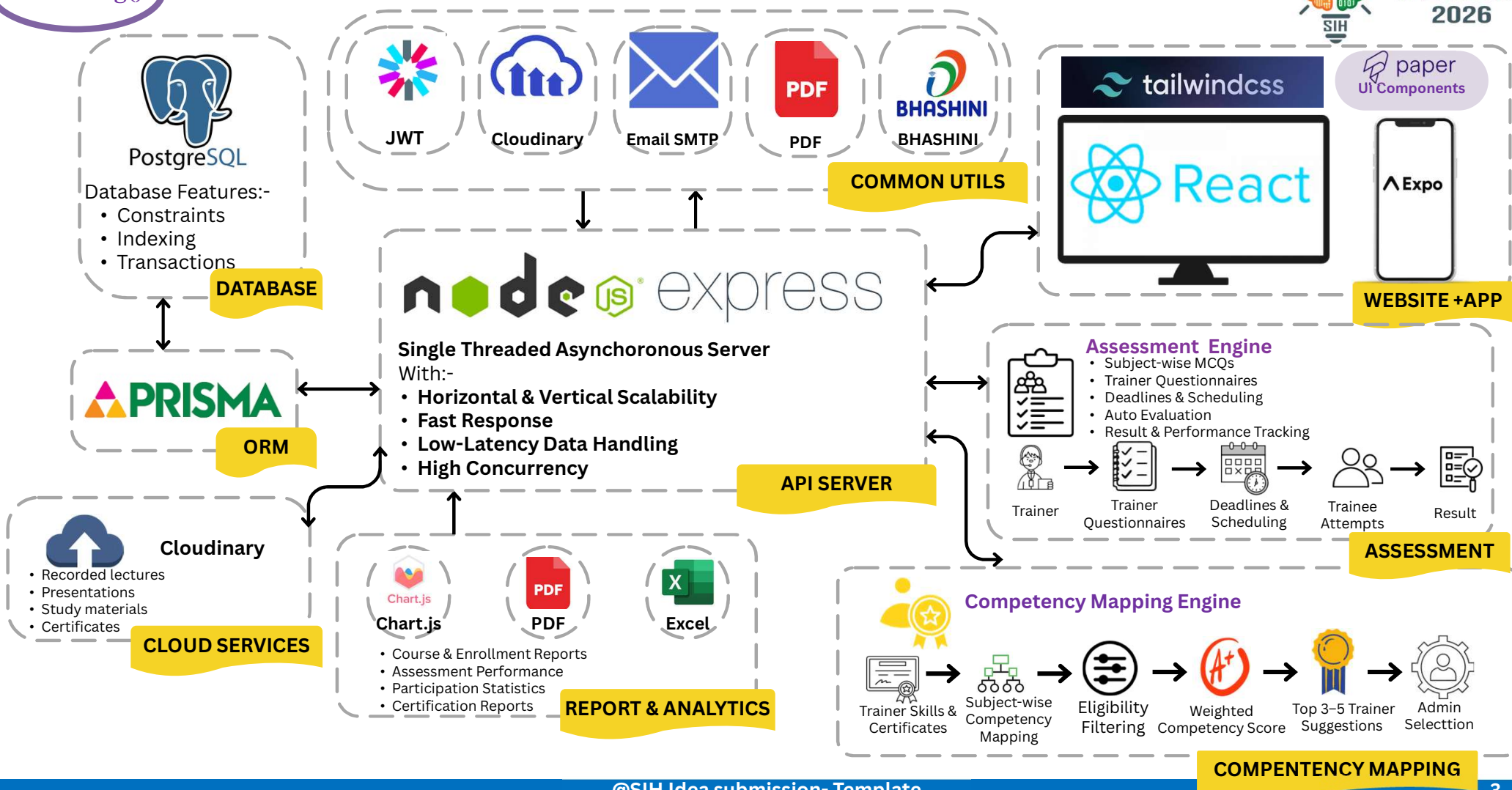
Connect • Learn • Assess • Analyse • Improve — all in one intelligent platform

ONE SOURCE OF TRUTH → SMARTER TRAINER MATCHING → FASTER LEARNING ACCESS → REAL-TIME PERFORMANCE VISIBILITY → BETTER TRAINING OUTCOMES

@SIH Idea submission- Template

TECHNICAL APPROACH

console.log()



CAPACITY CONNECT
MoES / IMD LMS Portal

Public Portal & Verify

Executive Dashboard

Course Catalog

Learning Gap Detection

AI Course Advisor

My Enrolled Courses

Scheduled Assessments

AI Practice Papers

Question Bank

Certified Credentials

Officer Profile

Competency Radar

Aniket Deshmukh
Scientific Assistant Grade-II

Sign Out

Exit Studio Radar Interpretation

Progress: 50%

Standard Split & Tools Full AI Summary Practice Quiz

VIDEO RESOURCE Unlocked & Accessible In-Portal Protected

Video 1: Recorded Masterclass — Sigma Coordinates & Lower Boundary Conditions

Uploaded by: Dr. Sunita Kulkarni • Uploaded on: Jan 15, 2025 • 45 mins

Mark Complete

Active Monsoon Conditions likely over northwest.

ISOLATED HEAVY TO VERY HEAVY RAINFALL AT FEW PLACES WITH EXTREMELY HEAVY RAINFALL LIKELY AT ISOLATED PLACES OVER JAMMU-KASHMIR AND HIMACHAL PRADESH ON 20TH & 21ST; OVER UTTARAKHAND, UTTAR PRADESH, BIHAR AND ASSAM & MEGHALAYA ON 20TH JULY.

DAILY WEATHER BRIEFING 20-07-2026 Watch on YouTube

Progression Rule: 80% Watch Threshold

PENDING (Requires > 80%) + 15% + 30%

Watched: 0% Unlock Target: 80% Fast-Forward to 85% (Unlock)

Video 1: Recorded Masterclass — Sigma Coordinates & Lower Boundary Conditions

MARKERS: 04:15

My Notes AI Summary Practice Quiz (AI)

GEMINI 1.5 FLASH Regenerate

AI Material Summary & Key Takeaways

EXECUTIVE SUMMARY

This curriculum resource provides comprehensive training on Numerical Weather Prediction (NWP) architectures, governing primitive equations, and high-resolution WRF model integration with atmospheric data assimilation systems.

Core Key Takeaways

- Formulation of primitive equations in hydrostatic and non-hydrostatic sigma-pressure vertical coordinate systems.
- Understanding Courant-Friedrichs-Lewy (CFL) numerical stability criteria for explicit and semi-implicit time-integration schemes.
- Parameterization of sub-grid scale processes: Planetary Boundary Layer (PBL) turbulence closures and microphysics schemes.
- Three-dimensional and four-dimensional variational data assimilation (3D-Var / 4D-Var) of satellite and radar observation vectors.

Governing Formulas & Theoretical Definitions

CFL Numerical Stability Condition: $C = (u \cdot \Delta t) / \Delta x \leq 1.0$ (governs spatial-temporal solver boundedness)

Hydrostatic Approximation: $\partial p / \partial z = -\rho g$ (valid for mesoscale to synoptic horizontal scales)

TARGET DIFFICULTY Medium (Operational)

QUESTION COUNT 5 Standard MCQs

Generate AI Quiz for "Video 1: Recorded Masterclass ..."

5 Questions Ready

1. In atmospheric models using the terrain-following Phillips sigma coordinate system (defined as $\sigma = p/p_s$), the horizontal pressure gradient force (PGF) is decomposed into two terms. What numerical challenge does this decomposition present over areas with highly complex, steep topography? Medium

A. The vertical advection term dominates the thermodynamic equation, leading to artificial gravity waves.

B. The two terms—the gradient of geopotential on the σ -surface and the surface pressure correction term—are often very large and of opposite signs, leading to massive truncation errors.

C. The surface pressure gradient becomes singular when the surface pressure p_s equals the reference pressure p_0 .

D. The coordinate transformation breaks down because the Jacobian of the transformation becomes zero at the surface boundary.

2. In the standard formulation of the Phillips sigma coordinate system ($\sigma = p/p_s$), what are the exact mathematical lower and upper boundary conditions for the vertical coordinate velocity $\dot{\sigma}$ assuming a flat or complex terrain surface with no mass flux across it? Medium



Potential Impact on Target Audience



Trainees

- Personalized learning & skill development
- Easier access to courses, assessments, certificates and feedback.



Trainers

- Better utilization of expertise
- Competency-based matching helps connect trainers with relevant training requirements.



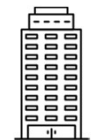
Administrators

- Data-driven training management
- Unified dashboards for participation, assessments, certifications and competencies.



Proven & Modular Implementation

- Core modules such as course management, assessments, user roles and dashboards can be implemented using established web technologies and a modular architecture.



High Organizational Demand

- Continuous training, competency development and knowledge sharing create a long-term need for a centralized capacity-building platform.



Benefits of the Solution



SOCIAL Inclusive & Accessible Learning

- Multilingual learning support
- Continuous professional development



ECONOMIC Efficient Training Utilization

- Reusable digital resources
- Better trainer-training alignment



ORGANIZATIONAL

- Competency gap identification
- Performance & feedback tracking



IMPACT



FEASIBILITY



Mature & Scalable Technology Stack

- React, Node.js, PostgreSQL and cloud services provide a reliable foundation for secure deployment and future system expansion.



VIABILITY



Sustainable & Scalable Growth

- Reusable learning content, competency tracking and cloud-based infrastructure enable sustainable operation and expansion across departments and organizations.

RESEARCH AND REFERENCES

1. iGOT Karmayogi – AI-Driven Capacity Building Plan

- Reference for role & competency mapping, competency-gap analysis, course recommendations and training analytics in government capacity building.
- Link:<https://portal.igotkarmayogi.gov.in/training-pla-ai/>

2. India Meteorological Department – Annual Report 2025

- Evidence of ongoing training and capacity-building programs, including integrated meteorological training, forecaster training and advanced training
- Link:https://metnet.imd.gov.in/docs/imdnews/ANNUAL_REPORT2025English.pdf

3. IMD – Capacity Building & Collaboration

- Documents IMD's formal training ecosystem, regional training centres, online/hybrid training and competency-oriented capacity building
- Link:https://internal.imd.gov.in/press_release/20250114_pr_3552.pdf

4. IMD – Time Evolution of Meteorological Training and Education in India

- Research describing IMD's training evolution and the use of competency requirements and blended learning in meteorological training.
- Link:<https://mausamjournal.imd.gov.in/index.php/MAUSAM>

5. Competency Management in Learning Management Systems

- Research on integrating competency mapping, evidence, proficiency assessment and competency reports into LMS platforms
- Link:<https://www.sciencedirect.com/science/article/pii/S187705091401103X>